Using R for Data Science Assignment

Asli Dagdelen

2024-11-27

R Markdown File for the Codes Used In this Assignment

All of the codes are taken from the class material unless stated otherwise.

The files containing sample annotation, expression data, gene annotation data and gene list data were copied from the source folder.

The data processing will be performed step by step. This file is containing codes that were used and the results of the codes while the report file includes the highlights of the codes and analysis of the graphs.

1. Getting the libraries that are being used in this assignment.

```
#This chunk of code is for the libraries that are going to be used in this assingment.
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.3     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(data.table)
```

```
##
## Attaching package: 'data.table'
##
## The following objects are masked from 'package:lubridate':
##
##     hour, isoweek, mday, minute, month, quarter, second, wday, week,
##     yday, year
##
## The following objects are masked from 'package:dplyr':
##
##     between, first, last
##
## The following object is masked from 'package:purrr':
##
##     transpose
```

```
library(pheatmap)
```

2. Loading the files that were copied from the source file.

```
#This chunk of code is for loading the files that are going to be used in this assingment into R.
sample_annot <- read.table("sample_annotation.csv", header=T, sep=",")
data_all <- read.table("data_all.csv", header=T, sep=",")
gene_annot <- read.table("gene_annotation.csv", header=T, sep=",")
#This line of code is to avoid getting the column X so that I do not have a duplicated column in the da
gene_annot_modified <- gene_annot[c("Type", "LongName")]
gene_list_table <- read.table("genelist_5.txt", header=T, sep=",")
```

3. Selecting the genes that were assigned to me in this coursework.

```
#This part of the code is for obtaining the data of choice.
my_genes <- data_all[c(61,64,104,10,39,69,112,60,28,52,35,175,102,89,135,190,57,111,185,37,71,62,182,165
                       121,42,183,88,76,105,23,158,9,198,54,82),]
#This part of the code is for obtaining the annotations for my data
annotations <- gene_annot_modified[c(61,64,104,10,39,69,112,60,28,52,35,175,102,89,135,190,57,111,185,37
                           121,42,183,88,76,105,23,158,9,198,54,82),]
```

4. Putting the data on a logaritmic scale

```
#This line of code is for getting the data on a log scale
expression <- log(my_genes[,2:13])
```

5. Binding the tables together to have a data frame that contains all of the information that is needed.

```
#This line is to bind the tables
table_bind <- cbind(expression, annotations)
#This line is to change the row name to the names of the genes
rownames(table_bind) <- table_bind$LongName
```

6. Using the table that was created, melt function was used to apply treatment groups on the data.

```
#This line of code is for creating a new data frame with table_bind dataframe so that melting does not
table_for_melting <- table_bind
#This line of code is for melting the data
#This line of code is taken from #https://stackoverflow.com/questions/2185252/reshaping-data-frame-from
table_for_melting <- melt(table_for_melting, id.vars=13:14, variable_name=c("A", "B", "C","D","E","F","
```

```
## Warning in melt.default(table_for_melting, id.vars = 13:14, variable_name =
## c("A", : The melt generic in data.table has been passed a data.frame and will
## attempt to redirect to the relevant reshape2 method; please note that reshape2
## is superseded and is no longer actively developed, and this redirection is now
## deprecated. To continue using melt methods from reshape2 while both libraries
## are attached, e.g. melt.list, you can prepend the namespace, i.e.
## reshape2::melt(table_for_melting). In the next version, this warning will
## become an error.
```

```
#These lines of code is to format the data frame into a data frame that has all the information and ann
treatment_groups <-as.data.frame(c(rep("1",120), rep("2",120), rep("3", 120), rep("4", 120)))
colnames(treatment_groups) <- "TreatmentGroup"
new_df_all <- cbind(table_for_melting, treatment_groups)
colnames(new_df_all)[2]<- "GeneName"
colnames(new_df_all)[3]<- "SampleName"
```

This shows an error for the function melt, however, the data frame is still created.

```
print(new_df_all)
```

```
##     Type                    GeneName SampleName     value TreatmentGroup
```

```
## 1     XA                  1421496_at          A 1.7347557             1
## 2     XA AFFX-18SRNAMur/X00686_M_at          A 2.2227087             1
## 3     XB                1438133_a_at          A 2.0926613             1
## 4     XA                1439766_x_at          A 1.6883511             1
## 5     XA                  1416157_at          A 1.7201533             1
## 6     XA                  1451620_at          A 1.1207913             1
## 7     XB                  1423824_at          A 1.3243176             1
## 8     XA                1430596_s_at          A 1.7076231             1
## 9     XA                1438511_a_at          A 1.3012264             1
## 10    XA                  1448022_at          A 1.2703582             1
## 11    XA                  1415983_at          A 1.8607925             1
## 12    XC                1420901_a_at          A 2.1059540             1
## 13    XB                  1452163_at          A 1.6240932             1
## 14    XB                1438069_a_at          A 1.6122304             1
## 15    XC                  1435397_at          A 1.7550157             1
## 16    XC                  1420998_at          A 2.2500669             1
## 17    XA                1456942_x_at          A 1.5405412             1
## 18    XB                  1438333_at          A 1.5144842             1
## 19    XC                  1418664_at          A 1.4438673             1
## 20    XA                1428653_x_at          A 0.9837037             1
## 21    XB                  1422458_at          A 2.2309173             1
## 22    XA                1431633_x_at          A 1.5879083             1
## 23    XC                  1453139_at          A 1.2390776             1
## 24    XC                1430640_a_at          A 1.6229756             1
## 25    XC                  1435595_at          A 1.0784515             1
## 26    XA                  1415938_at          A 2.0907495             1
## 27    XA AFFX-18SRNAMur/X00686_5_at          A 2.3548659             1
## 28    XA                1421100_a_at          A 1.2985243             1
## 29    XB                1456475_s_at          A 1.7271795             1
## 30    XA                  1455169_at          A 1.2284724             1
## 31    XC                1451809_s_at          A 1.3454822             1
## 32    XB                  1438664_at          A 1.6818822             1
## 33    XB                1441667_s_at          A 1.5253290             1
## 34    XB                  1417956_at          A 1.7555863             1
## 35    XA                  1432227_at          A 1.7278696             1
## 36    XC                  1438883_at          A 1.3555003             1
## 37    XA                1436948_a_at          A 0.9190906             1
## 38    XC                1436959_x_at          A 1.6447440             1
## 39    XA                  1416418_at          A 1.8364192             1
## 40    XB                  1435436_at          A 2.1156501             1
## 41    XA                  1421496_at          B 1.7605579             1
## 42    XA AFFX-18SRNAMur/X00686_M_at          B 1.9394591             1
## 43    XB                1438133_a_at          B 2.1191607             1
## 44    XA                1439766_x_at          B 1.7048565             1
## 45    XA                  1416157_at          B 1.6140786             1
## 46    XA                  1451620_at          B 1.2917416             1
## 47    XB                  1423824_at          B 1.2736961             1
## 48    XA                1430596_s_at          B 1.7260779             1
## 49    XA                1438511_a_at          B 1.2739636             1
## 50    XA                  1448022_at          B 1.1120381             1
## 51    XA                  1415983_at          B 1.8992470             1
## 52    XC                1420901_a_at          B 2.1042925             1
## 53    XB                  1452163_at          B 1.6461957             1
## 54    XB                1438069_a_at          B 1.3306323             1
```

```
## 55    XC                  1435397_at          B 1.7805966             1
## 56    XC                  1420998_at          B 2.2129657             1
## 57    XA                1456942_x_at          B 1.4966423             1
## 58    XB                  1438333_at          B 1.4251719             1
## 59    XC                  1418664_at          B 1.4884948             1
## 60    XA                1428653_x_at          B 1.0714496             1
## 61    XB                  1422458_at          B 2.2210954             1
## 62    XA                1431633_x_at          B 1.5436159             1
## 63    XC                  1453139_at          B 1.2932499             1
## 64    XC                1430640_a_at          B 1.6322374             1
## 65    XC                  1435595_at          B 0.9651065             1
## 66    XA                  1415938_at          B 2.0643861             1
## 67    XA AFFX-18SRNAMur/X00686_5_at          B 2.0267782             1
## 68    XA                1421100_a_at          B 1.3324913             1
## 69    XB                1456475_s_at          B 1.7805907             1
## 70    XA                  1455169_at          B 1.2975788             1
## 71    XC                1451809_s_at          B 1.3753088             1
## 72    XB                  1438664_at          B 1.7109680             1
## 73    XB                1441667_s_at          B 1.4804269             1
## 74    XB                  1417956_at          B 1.6883786             1
## 75    XA                  1432227_at          B 1.7175373             1
## 76    XC                  1438883_at          B 1.5392474             1
## 77    XA                1436948_a_at          B 1.1067413             1
## 78    XC                1436959_x_at          B 1.5941130             1
## 79    XA                  1416418_at          B 1.8854717             1
## 80    XB                  1435436_at          B 2.1413410             1
## 81    XA                  1421496_at          C 1.8198205             1
## 82    XA AFFX-18SRNAMur/X00686_M_at          C 2.2128087             1
## 83    XB                1438133_a_at          C 2.1304262             1
## 84    XA                1439766_x_at          C 1.6976415             1
## 85    XA                  1416157_at          C 1.8196438             1
## 86    XA                  1451620_at          C 1.2185113             1
## 87    XB                  1423824_at          C 1.2854160             1
## 88    XA                1430596_s_at          C 1.7732817             1
## 89    XA                1438511_a_at          C 1.2877375             1
## 90    XA                  1448022_at          C 1.4138636             1
## 91    XA                  1415983_at          C 1.8904319             1
## 92    XC                1420901_a_at          C 2.1102938             1
## 93    XB                  1452163_at          C 1.6504884             1
## 94    XB                1438069_a_at          C 1.6394335             1
## 95    XC                  1435397_at          C 1.8099720             1
## 96    XC                  1420998_at          C 2.2684371             1
## 97    XA                1456942_x_at          C 1.6256465             1
## 98    XB                  1438333_at          C 1.4558240             1
## 99    XC                  1418664_at          C 1.5605750             1
## 100   XA                1428653_x_at          C 1.1212803             1
## 101   XB                  1422458_at          C 2.2456886             1
## 102   XA                1431633_x_at          C 1.6225344             1
## 103   XC                  1453139_at          C 1.2777161             1
## 104   XC                1430640_a_at          C 1.6513782             1
## 105   XC                  1435595_at          C 0.9461276             1
## 106   XA                  1415938_at          C 2.1172453             1
## 107   XA AFFX-18SRNAMur/X00686_5_at          C 2.3371635             1
## 108   XA                1421100_a_at          C 1.3342549             1
```

```
## 109   XB               1456475_s_at         C 1.8009445             1
## 110   XA                 1455169_at         C 1.2462533             1
## 111   XC               1451809_s_at         C 1.3953136             1
## 112   XB                 1438664_at         C 1.7102750             1
## 113   XB               1441667_s_at         C 1.5495437             1
## 114   XB                 1417956_at         C 1.7258088             1
## 115   XA                 1432227_at         C 1.7301583             1
## 116   XC                 1438883_at         C 1.4604812             1
## 117   XA               1436948_a_at         C 0.9678990             1
## 118   XC               1436959_x_at         C 1.6548321             1
## 119   XA                 1416418_at         C 1.8412304             1
## 120   XB                 1435436_at         C 2.1319525             1
## 121   XA                 1421496_at         D 1.4037561             2
## 122   XA AFFX-18SRNAMur/X00686_M_at         D 2.1593669             2
## 123   XB               1438133_a_at         D 1.5829215             2
## 124   XA               1439766_x_at         D 1.7151519             2
## 125   XA                 1416157_at         D 1.2238696             2
## 126   XA                 1451620_at         D 1.1821668             2
## 127   XB                 1423824_at         D 1.2708943             2
## 128   XA               1430596_s_at         D 1.6211812             2
## 129   XA               1438511_a_at         D 1.4377457             2
## 130   XA                 1448022_at         D 0.9258273             2
## 131   XA                 1415983_at         D 1.4546414             2
## 132   XC               1420901_a_at         D 1.6272695             2
## 133   XB                 1452163_at         D 1.3800858             2
## 134   XB               1438069_a_at         D 1.0910499             2
## 135   XC                 1435397_at         D 1.8181219             2
## 136   XC                 1420998_at         D 1.9645800             2
## 137   XA               1456942_x_at         D 1.4498702             2
## 138   XB                 1438333_at         D 1.1735412             2
## 139   XC                 1418664_at         D 1.6766274             2
## 140   XA               1428653_x_at         D 1.0413755             2
## 141   XB                 1422458_at         D 2.2119076             2
## 142   XA               1431633_x_at         D 1.7115261             2
## 143   XC                 1453139_at         D 1.4256217             2
## 144   XC               1430640_a_at         D 1.6333432             2
## 145   XC                 1435595_at         D 1.1395870             2
## 146   XA                 1415938_at         D 2.1593282             2
## 147   XA AFFX-18SRNAMur/X00686_5_at         D 2.2621227             2
## 148   XA               1421100_a_at         D 1.5123200             2
## 149   XB               1456475_s_at         D 1.7889543             2
## 150   XA                 1455169_at         D 1.3033127             2
## 151   XC               1451809_s_at         D 1.3302977             2
## 152   XB                 1438664_at         D 1.7319239             2
## 153   XB               1441667_s_at         D 1.1216179             2
## 154   XB                 1417956_at         D 1.6514464             2
## 155   XA                 1432227_at         D 1.7110476             2
## 156   XC                 1438883_at         D 1.7119610             2
## 157   XA               1436948_a_at         D 1.1229213             2
## 158   XC               1436959_x_at         D 1.7198405             2
## 159   XA                 1416418_at         D 1.3189132             2
## 160   XB                 1435436_at         D 2.1453312             2
## 161   XA                 1421496_at         E 1.4823529             2
## 162   XA AFFX-18SRNAMur/X00686_M_at         E 2.1696332             2
```

```
## 163   XB                1438133_a_at          E 1.7766631              2
## 164   XA                1439766_x_at          E 1.6948617              2
## 165   XA                  1416157_at          E 1.2999383              2
## 166   XA                  1451620_at          E 1.2223097              2
## 167   XB                  1423824_at          E 1.2889208              2
## 168   XA                1430596_s_at          E 1.5815638              2
## 169   XA                1438511_a_at          E 1.3327725              2
## 170   XA                  1448022_at          E 0.7922652              2
## 171   XA                  1415983_at          E 1.3590033              2
## 172   XC                1420901_a_at          E 1.7925915              2
## 173   XB                  1452163_at          E 1.3398524              2
## 174   XB                1438069_a_at          E 1.1937363              2
## 175   XC                  1435397_at          E 1.7921077              2
## 176   XC                  1420998_at          E 2.0469482              2
## 177   XA                1456942_x_at          E 1.4630611              2
## 178   XB                  1438333_at          E 1.2427718              2
## 179   XC                  1418664_at          E 1.6401838              2
## 180   XA                1428653_x_at          E 1.0116879              2
## 181   XB                  1422458_at          E 2.2615370              2
## 182   XA                1431633_x_at          E 1.7765376              2
## 183   XC                  1453139_at          E 1.3729248              2
## 184   XC                1430640_a_at          E 1.6413646              2
## 185   XC                  1435595_at          E 1.1059830              2
## 186   XA                  1415938_at          E 2.1656376              2
## 187   XA AFFX-18SRNAMur/X00686_5_at          E 2.2569618              2
## 188   XA                1421100_a_at          E 1.5842850              2
## 189   XB                1456475_s_at          E 1.7863640              2
## 190   XA                  1455169_at          E 1.2613690              2
## 191   XC                1451809_s_at          E 1.3480765              2
## 192   XB                  1438664_at          E 1.6469314              2
## 193   XB                1441667_s_at          E 1.3625494              2
## 194   XB                  1417956_at          E 1.7600178              2
## 195   XA                  1432227_at          E 1.7789467              2
## 196   XC                  1438883_at          E 1.6703404              2
## 197   XA                1436948_a_at          E 1.0236633              2
## 198   XC                1436959_x_at          E 1.7421975              2
## 199   XA                  1416418_at          E 1.4982559              2
## 200   XB                  1435436_at          E 2.1808604              2
## 201   XA                  1421496_at          F 1.4144084              2
## 202   XA AFFX-18SRNAMur/X00686_M_at          F 2.3110101              2
## 203   XB                1438133_a_at          F 1.7042039              2
## 204   XA                1439766_x_at          F 1.7191198              2
## 205   XA                  1416157_at          F 1.3474067              2
## 206   XA                  1451620_at          F 1.2922829              2
## 207   XB                  1423824_at          F 1.3234819              2
## 208   XA                1430596_s_at          F 1.6496274              2
## 209   XA                1438511_a_at          F 1.4279460              2
## 210   XA                  1448022_at          F 0.8102545              2
## 211   XA                  1415983_at          F 1.3982290              2
## 212   XC                1420901_a_at          F 1.7474743              2
## 213   XB                  1452163_at          F 1.3384979              2
## 214   XB                1438069_a_at          F 1.1424422              2
## 215   XC                  1435397_at          F 1.8108665              2
## 216   XC                  1420998_at          F 2.0139485              2
```

```
## 217   XA                1456942_x_at          F 1.4843308             2
## 218   XB                  1438333_at          F 1.0730667             2
## 219   XC                  1418664_at          F 1.6490073             2
## 220   XA                1428653_x_at          F 0.9998923             2
## 221   XB                  1422458_at          F 2.2374745             2
## 222   XA                1431633_x_at          F 1.7966420             2
## 223   XC                  1453139_at          F 1.4530201             2
## 224   XC                1430640_a_at          F 1.6379180             2
## 225   XC                  1435595_at          F 1.1015603             2
## 226   XA                  1415938_at          F 2.1664638             2
## 227   XA AFFX-18SRNAMur/X00686_5_at          F 2.3862640             2
## 228   XA                1421100_a_at          F 1.5428994             2
## 229   XB                1456475_s_at          F 1.7981780             2
## 230   XA                  1455169_at          F 1.2857613             2
## 231   XC                1451809_s_at          F 1.3449328             2
## 232   XB                  1438664_at          F 1.6869245             2
## 233   XB                1441667_s_at          F 1.1738947             2
## 234   XB                  1417956_at          F 1.7154366             2
## 235   XA                  1432227_at          F 1.7921105             2
## 236   XC                  1438883_at          F 1.7076549             2
## 237   XA                1436948_a_at          F 1.0036649             2
## 238   XC                1436959_x_at          F 1.7714677             2
## 239   XA                  1416418_at          F 1.3359713             2
## 240   XB                  1435436_at          F 2.1417704             2
## 241   XA                  1421496_at          G 1.3554150             3
## 242   XA AFFX-18SRNAMur/X00686_M_at          G 2.1985274             3
## 243   XB                1438133_a_at          G 1.6903044             3
## 244   XA                1439766_x_at          G 1.4096273             3
## 245   XA                  1416157_at          G 1.2770762             3
## 246   XA                  1451620_at          G 1.2468187             3
## 247   XB                  1423824_at          G 1.2859061             3
## 248   XA                1430596_s_at          G 1.4862806             3
## 249   XA                1438511_a_at          G 1.4991182             3
## 250   XA                  1448022_at          G 0.9288828             3
## 251   XA                  1415983_at          G 1.4599620             3
## 252   XC                1420901_a_at          G 1.8079208             3
## 253   XB                  1452163_at          G 1.5201373             3
## 254   XB                1438069_a_at          G 1.2306832             3
## 255   XC                  1435397_at          G 1.8124837             3
## 256   XC                  1420998_at          G 1.9367976             3
## 257   XA                1456942_x_at          G 1.5995280             3
## 258   XB                  1438333_at          G 1.3180529             3
## 259   XC                  1418664_at          G 1.4968275             3
## 260   XA                1428653_x_at          G 1.0536735             3
## 261   XB                  1422458_at          G 2.2104383             3
## 262   XA                1431633_x_at          G 1.7978749             3
## 263   XC                  1453139_at          G 1.4811634             3
## 264   XC                1430640_a_at          G 1.6669187             3
## 265   XC                  1435595_at          G 1.1697145             3
## 266   XA                  1415938_at          G 2.0842287             3
## 267   XA AFFX-18SRNAMur/X00686_5_at          G 2.2611602             3
## 268   XA                1421100_a_at          G 1.5220878             3
## 269   XB                1456475_s_at          G 1.8057188             3
## 270   XA                  1455169_at          G 1.2401085             3
```

```
## 271   XC               1451809_s_at         G 1.3216697             3
## 272   XB                 1438664_at         G 1.6842136             3
## 273   XB               1441667_s_at         G 1.1424949             3
## 274   XB                 1417956_at         G 1.8583482             3
## 275   XA                 1432227_at         G 1.4459941             3
## 276   XC                 1438883_at         G 1.6975674             3
## 277   XA               1436948_a_at         G 0.9733095             3
## 278   XC               1436959_x_at         G 1.8324308             3
## 279   XA                 1416418_at         G 1.5088221             3
## 280   XB                 1435436_at         G 1.9975982             3
## 281   XA                 1421496_at         H 1.4115299             3
## 282   XA AFFX-18SRNAMur/X00686_M_at         H 2.2391579             3
## 283   XB               1438133_a_at         H 1.7929595             3
## 284   XA               1439766_x_at         H 1.2997842             3
## 285   XA                 1416157_at         H 1.3086845             3
## 286   XA                 1451620_at         H 1.2344684             3
## 287   XB                 1423824_at         H 1.3778627             3
## 288   XA               1430596_s_at         H 1.4187145             3
## 289   XA               1438511_a_at         H 1.4288032             3
## 290   XA                 1448022_at         H 0.8722569             3
## 291   XA                 1415983_at         H 1.4162147             3
## 292   XC               1420901_a_at         H 1.8518510             3
## 293   XB                 1452163_at         H 1.4959037             3
## 294   XB               1438069_a_at         H 1.1750579             3
## 295   XC                 1435397_at         H 1.8472229             3
## 296   XC                 1420998_at         H 2.0233443             3
## 297   XA               1456942_x_at         H 1.4320526             3
## 298   XB                 1438333_at         H 1.3289063             3
## 299   XC                 1418664_at         H 1.6311877             3
## 300   XA               1428653_x_at         H 0.9417890             3
## 301   XB                 1422458_at         H 2.1995944             3
## 302   XA               1431633_x_at         H 1.6987427             3
## 303   XC                 1453139_at         H 1.3198810             3
## 304   XC               1430640_a_at         H 1.6423699             3
## 305   XC                 1435595_at         H 1.2010214             3
## 306   XA                 1415938_at         H 2.0822626             3
## 307   XA AFFX-18SRNAMur/X00686_5_at         H 2.3540586             3
## 308   XA               1421100_a_at         H 1.4203316             3
## 309   XB               1456475_s_at         H 1.7387387             3
## 310   XA                 1455169_at         H 1.2886153             3
## 311   XC               1451809_s_at         H 1.3291366             3
## 312   XB                 1438664_at         H 1.6914791             3
## 313   XB               1441667_s_at         H 1.2924089             3
## 314   XB                 1417956_at         H 1.8523322             3
## 315   XA                 1432227_at         H 1.4876939             3
## 316   XC                 1438883_at         H 1.7392646             3
## 317   XA               1436948_a_at         H 0.8488630             3
## 318   XC               1436959_x_at         H 1.7334880             3
## 319   XA                 1416418_at         H 1.4176691             3
## 320   XB                 1435436_at         H 1.9788992             3
## 321   XA                 1421496_at         I 1.4207871             3
## 322   XA AFFX-18SRNAMur/X00686_M_at         I 2.2943063             3
## 323   XB               1438133_a_at         I 1.8388005             3
## 324   XA               1439766_x_at         I 1.3799758             3
```

```
## 325   XA                  1416157_at          I 1.3090927             3
## 326   XA                  1451620_at          I 1.2111933             3
## 327   XB                  1423824_at          I 1.3506795             3
## 328   XA                 1430596_s_at         I 1.4191643             3
## 329   XA                 1438511_a_at         I 1.4701411             3
## 330   XA                  1448022_at          I 0.8076435             3
## 331   XA                  1415983_at          I 1.4948220             3
## 332   XC                 1420901_a_at         I 1.8017032             3
## 333   XB                  1452163_at          I 1.5277398             3
## 334   XB                 1438069_a_at         I 1.2379351             3
## 335   XC                  1435397_at          I 1.8498187             3
## 336   XC                  1420998_at          I 2.0210601             3
## 337   XA                 1456942_x_at         I 1.4910886             3
## 338   XB                  1438333_at          I 1.4048215             3
## 339   XC                  1418664_at          I 1.5981043             3
## 340   XA                 1428653_x_at         I 1.0445938             3
## 341   XB                  1422458_at          I 2.2065851             3
## 342   XA                 1431633_x_at         I 1.7532604             3
## 343   XC                  1453139_at          I 1.4613366             3
## 344   XC                 1430640_a_at         I 1.6677647             3
## 345   XC                  1435595_at          I 1.1638420             3
## 346   XA                  1415938_at          I 2.1358749             3
## 347   XA AFFX-18SRNAMur/X00686_5_at          I 2.3860868             3
## 348   XA                 1421100_a_at         I 1.4559266             3
## 349   XB                 1456475_s_at         I 1.8442174             3
## 350   XA                  1455169_at          I 1.2854160             3
## 351   XC                 1451809_s_at         I 1.3446344             3
## 352   XB                  1438664_at          I 1.7344718             3
## 353   XB                 1441667_s_at         I 1.2910066             3
## 354   XB                  1417956_at          I 1.8323087             3
## 355   XA                  1432227_at          I 1.4912212             3
## 356   XC                  1438883_at          I 1.8157468             3
## 357   XA                 1436948_a_at         I 1.0179491             3
## 358   XC                 1436959_x_at         I 1.8036997             3
## 359   XA                  1416418_at          I 1.5076413             3
## 360   XB                  1435436_at          I 2.0322770             3
## 361   XA                  1421496_at          J 1.3136946             4
## 362   XA AFFX-18SRNAMur/X00686_M_at          J 1.8250769             4
## 363   XB                 1438133_a_at         J 1.8039654             4
## 364   XA                 1439766_x_at         J 1.1026593             4
## 365   XA                  1416157_at          J 1.2527693             4
## 366   XA                  1451620_at          J 1.5704244             4
## 367   XB                  1423824_at          J 1.7104762             4
## 368   XA                 1430596_s_at         J 1.9259779             4
## 369   XA                 1438511_a_at         J 1.8087825             4
## 370   XA                  1448022_at          J 1.2045738             4
## 371   XA                  1415983_at          J 1.7038243             4
## 372   XC                 1420901_a_at         J 1.7921174             4
## 373   XB                  1452163_at          J 1.7878305             4
## 374   XB                 1438069_a_at         J 1.1671241             4
## 375   XC                  1435397_at          J 2.1807030             4
## 376   XC                  1420998_at          J 1.7820132             4
## 377   XA                 1456942_x_at         J 1.9115590             4
## 378   XB                  1438333_at          J 1.6959255             4
```

```
## 379   XC                 1418664_at          J 1.8746791             4
## 380   XA               1428653_x_at          J 1.4252672             4
## 381   XB                 1422458_at          J 1.7767116             4
## 382   XA               1431633_x_at          J 1.1908005             4
## 383   XC                 1453139_at          J 1.7062009             4
## 384   XC               1430640_a_at          J 2.0142291             4
## 385   XC                 1435595_at          J 1.2182776             4
## 386   XA                 1415938_at          J 1.6760406             4
## 387   XA AFFX-18SRNAMur/X00686_5_at          J 1.8724015             4
## 388   XA               1421100_a_at          J 1.0257611             4
## 389   XB               1456475_s_at          J 2.1609771             4
## 390   XA                 1455169_at          J 1.7467283             4
## 391   XC               1451809_s_at          J 1.6790308             4
## 392   XB                 1438664_at          J 2.1021288             4
## 393   XB               1441667_s_at          J 1.5032232             4
## 394   XB                 1417956_at          J 2.0914961             4
## 395   XA                 1432227_at          J 1.4047550             4
## 396   XC                 1438883_at          J 1.3942749             4
## 397   XA               1436948_a_at          J 1.5846263             4
## 398   XC               1436959_x_at          J 2.0358467             4
## 399   XA                 1416418_at          J 1.5846390             4
## 400   XB                 1435436_at          J 1.6864999             4
## 401   XA                 1421496_at          K 1.2586431             4
## 402   XA AFFX-18SRNAMur/X00686_M_at          K 1.9917337             4
## 403   XB               1438133_a_at          K 1.8769845             4
## 404   XA               1439766_x_at          K 0.9775801             4
## 405   XA                 1416157_at          K 1.2904036             4
## 406   XA                 1451620_at          K 1.6548025             4
## 407   XB                 1423824_at          K 1.6337579             4
## 408   XA               1430596_s_at          K 1.9618640             4
## 409   XA               1438511_a_at          K 1.8506956             4
## 410   XA                 1448022_at          K 1.0854613             4
## 411   XA                 1415983_at          K 1.7861158             4
## 412   XC               1420901_a_at          K 1.8059201             4
## 413   XB                 1452163_at          K 1.7863350             4
## 414   XB               1438069_a_at          K 1.2138740             4
## 415   XC                 1435397_at          K 2.1468963             4
## 416   XC                 1420998_at          K 1.8827247             4
## 417   XA               1456942_x_at          K 1.9273056             4
## 418   XB                 1438333_at          K 1.6166676             4
## 419   XC                 1418664_at          K 1.8574538             4
## 420   XA               1428653_x_at          K 1.4385810             4
## 421   XB                 1422458_at          K 1.8150985             4
## 422   XA               1431633_x_at          K 1.3708464             4
## 423   XC                 1453139_at          K 1.6375791             4
## 424   XC               1430640_a_at          K 1.9992660             4
## 425   XC                 1435595_at          K 1.4501003             4
## 426   XA                 1415938_at          K 1.7019432             4
## 427   XA AFFX-18SRNAMur/X00686_5_at          K 2.1781636             4
## 428   XA               1421100_a_at          K 0.9529047             4
## 429   XB               1456475_s_at          K 2.1611059             4
## 430   XA                 1455169_at          K 1.6095894             4
## 431   XC               1451809_s_at          K 1.6874728             4
## 432   XB                 1438664_at          K 2.0726313             4
```

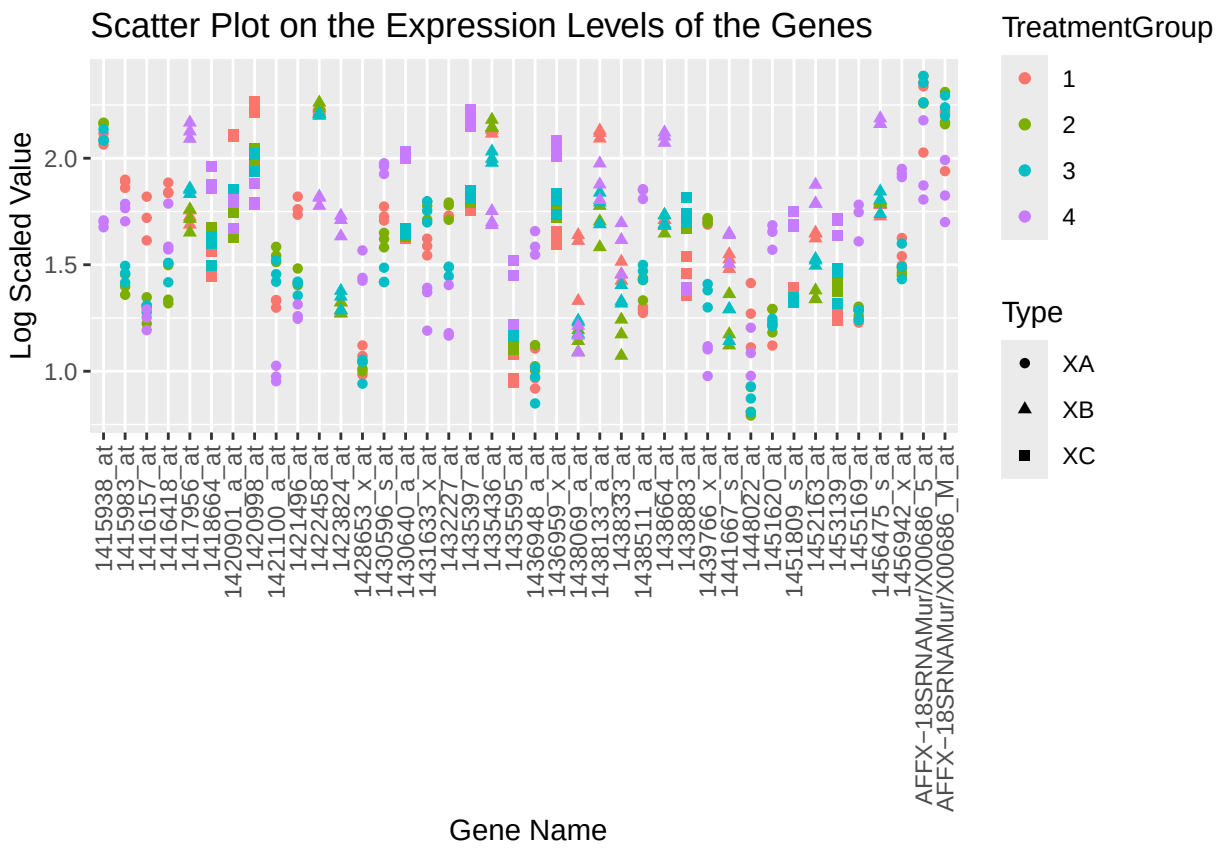
```
## 433   XB               1441667_s_at          K 1.6470634             4
## 434   XB                 1417956_at          K 2.1675880             4
## 435   XA                 1432227_at          K 1.1788912             4
## 436   XC                 1438883_at          K 1.3800050             4
## 437   XA               1436948_a_at          K 1.5471161             4
## 438   XC               1436959_x_at          K 2.0080823             4
## 439   XA                 1416418_at          K 1.7872144             4
## 440   XB                 1435436_at          K 1.6991745             4
## 441   XA                 1421496_at          L 1.2465866             4
## 442   XA AFFX-18SRNAMur/X00686_M_at          L 1.7001679             4
## 443   XB               1438133_a_at          L 1.9758693             4
## 444   XA               1439766_x_at          L 1.1171626             4
## 445   XA                 1416157_at          L 1.1930042             4
## 446   XA                 1451620_at          L 1.6852319             4
## 447   XB                 1423824_at          L 1.7302608             4
## 448   XA               1430596_s_at          L 1.9764045             4
## 449   XA               1438511_a_at          L 1.8548758             4
## 450   XA                 1448022_at          L 0.9785683             4
## 451   XA                 1415983_at          L 1.7654200             4
## 452   XC               1420901_a_at          L 1.6720080             4
## 453   XB                 1452163_at          L 1.8756307             4
## 454   XB               1438069_a_at          L 1.0877309             4
## 455   XC                 1435397_at          L 2.2281763             4
## 456   XC                 1420998_at          L 1.7910014             4
## 457   XA               1456942_x_at          L 1.9499015             4
## 458   XB                 1438333_at          L 1.4540337             4
## 459   XC                 1418664_at          L 1.9610299             4
## 460   XA               1428653_x_at          L 1.5674345             4
## 461   XB                 1422458_at          L 1.8200792             4
## 462   XA               1431633_x_at          L 1.3916298             4
## 463   XC                 1453139_at          L 1.7155538             4
## 464   XC               1430640_a_at          L 2.0310537             4
## 465   XC                 1435595_at          L 1.5183786             4
## 466   XA                 1415938_at          L 1.7080729             4
## 467   XA AFFX-18SRNAMur/X00686_5_at          L 1.8060793             4
## 468   XA               1421100_a_at          L 0.9750036             4
## 469   XB               1456475_s_at          L 2.1874607             4
## 470   XA                 1455169_at          L 1.7819106             4
## 471   XC               1451809_s_at          L 1.7493756             4
## 472   XB                 1438664_at          L 2.1213160             4
## 473   XB               1441667_s_at          L 1.6398242             4
## 474   XB                 1417956_at          L 2.1259937             4
## 475   XA                 1432227_at          L 1.1674314             4
## 476   XC                 1438883_at          L 1.3823378             4
## 477   XA               1436948_a_at          L 1.6577535             4
## 478   XC               1436959_x_at          L 2.0839861             4
## 479   XA                 1416418_at          L 1.5719221             4
## 480   XB                 1435436_at          L 1.7525455             4
```

Part A: Scatter Plot and Box Plot

Melted table was used to create the scatter plot and the box plot graphs.

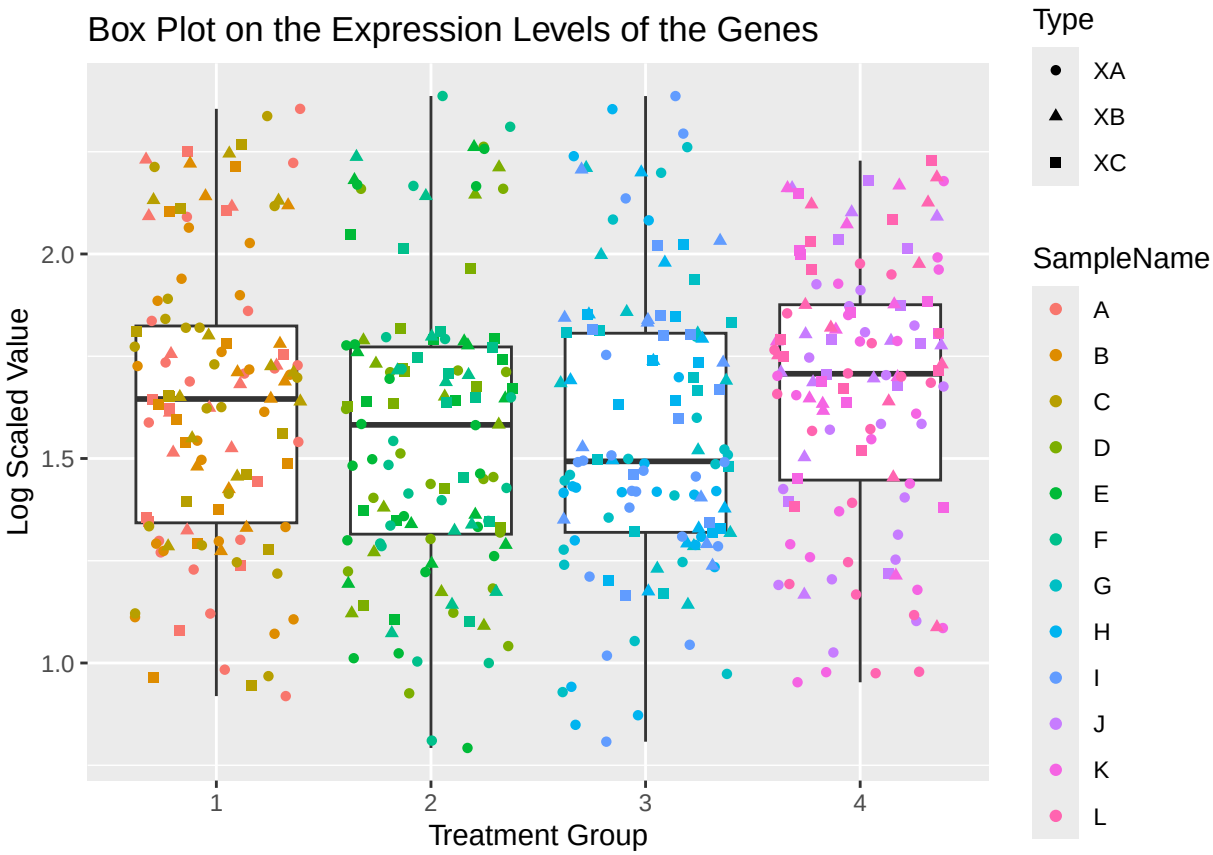
7. After this table is created, a scatter plot of the gene names against their values was generated. The types of the genes were assigned to shapes while treatment groups were assigned to colours.

```
#This chunk of code is to plot a graph (Gene Name against value)
#This chunk of code was created with the help of codes given in class and https://ggplot2-book.org/anno
ggplot(data = new_df_all) +
  geom_point(aes(y = value, x=GeneName, colour = TreatmentGroup, shape=Type)) +
  labs(y = "Log Scaled Value", x = " Gene Name",
       title="Scatter Plot on the Expression Levels of the Genes")+
  #This part of the code was taken from https://stackoverflow.com/questions/1330989/rotating-and-spacin
  #to change the angle of the Gene Names on the X axis
  scale_x_discrete(guide = guide_axis(angle = 90))
```



8. Using the same data frame, four box plot graphs were created according to the treatment groups. Type of the genes are labeled as shapes and sample names are labeled with colours.

```
#This chunk of code was created with the help of codes given in class and https://ggplot2-book.org/anno
ggplot(data = new_df_all) +
  geom_boxplot(aes(y=value,x=TreatmentGroup))+
  geom_jitter(aes(y=value, x=TreatmentGroup,shape =Type,colour=SampleName))+
  labs(y = "Log Scaled Value", x = "Treatment Group",
       title="Box Plot on the Expression Levels of the Genes")
```



Part B: Heatmaps

Using the table_bind data frame, 2 heatmaps were created: one with only genes clustered and another one with genes and samples clustered.

9. To create informative heatmaps, annotations are needed. Hence, two data frames including row names and column names for annotation.

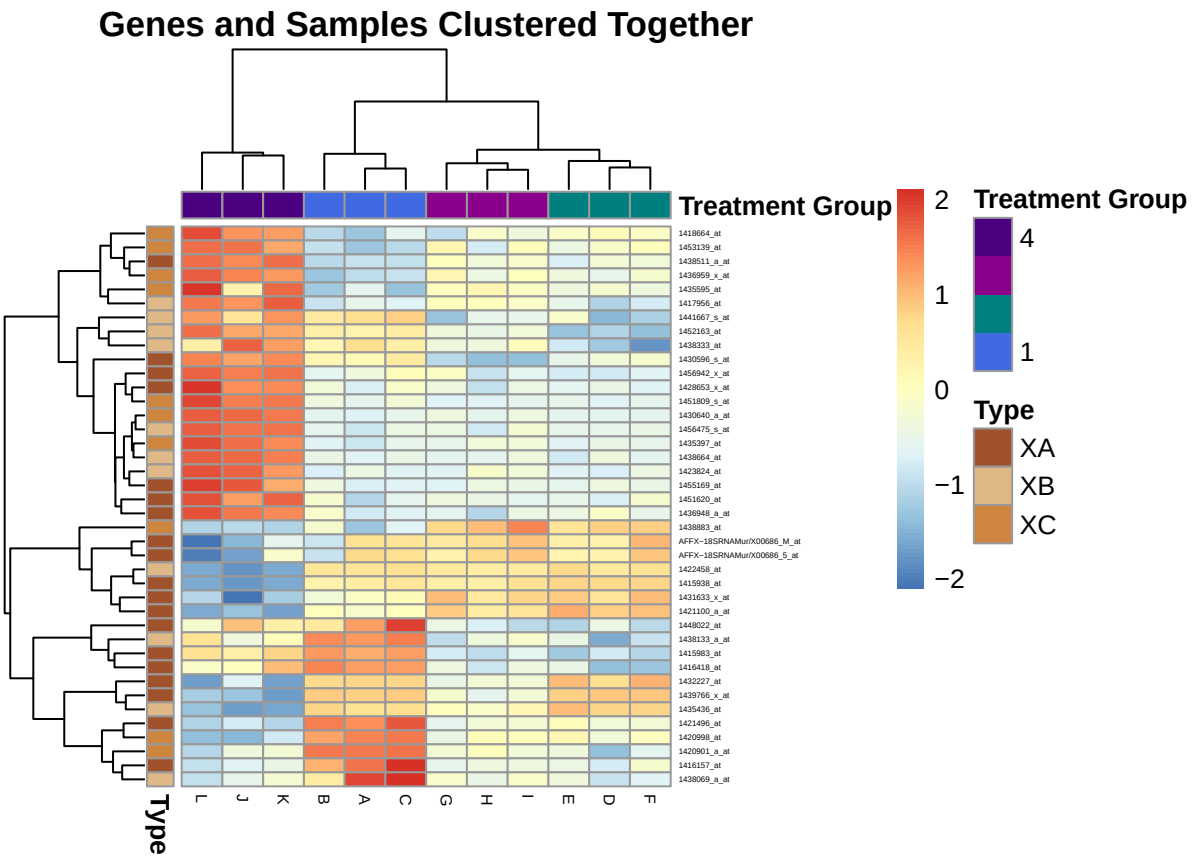
```
#These lines of code was created for the row and column names of annotation.
#For row names:
rows_heatmap <- as.data.frame(table_bind$Type)
row.names(rows_heatmap)<- table_bind$LongName
colnames(rows_heatmap)<-"Type"
#For column names:
cols_heatmap <- as.data.frame(sample_annot$TreatmentGroup)
row.names(cols_heatmap) <- sample_annot$SampleName
colnames(cols_heatmap)<- "Treatment Group"
```

Then, a list of colours were assigned to each value of the rows and columns.

```
#This list is created for the colour assignment.
annotation_list=list("Treatment Group" = c("1"="#4169E1",
                                           "2"="#008080",
                                           "3"="#8B008B",
                                           "4"="#4B0082"),
                     "Type" = c("XA"="#A0522D",
                                "XB"="#DEB887",
                                "XC"="#CD853F" ))
```

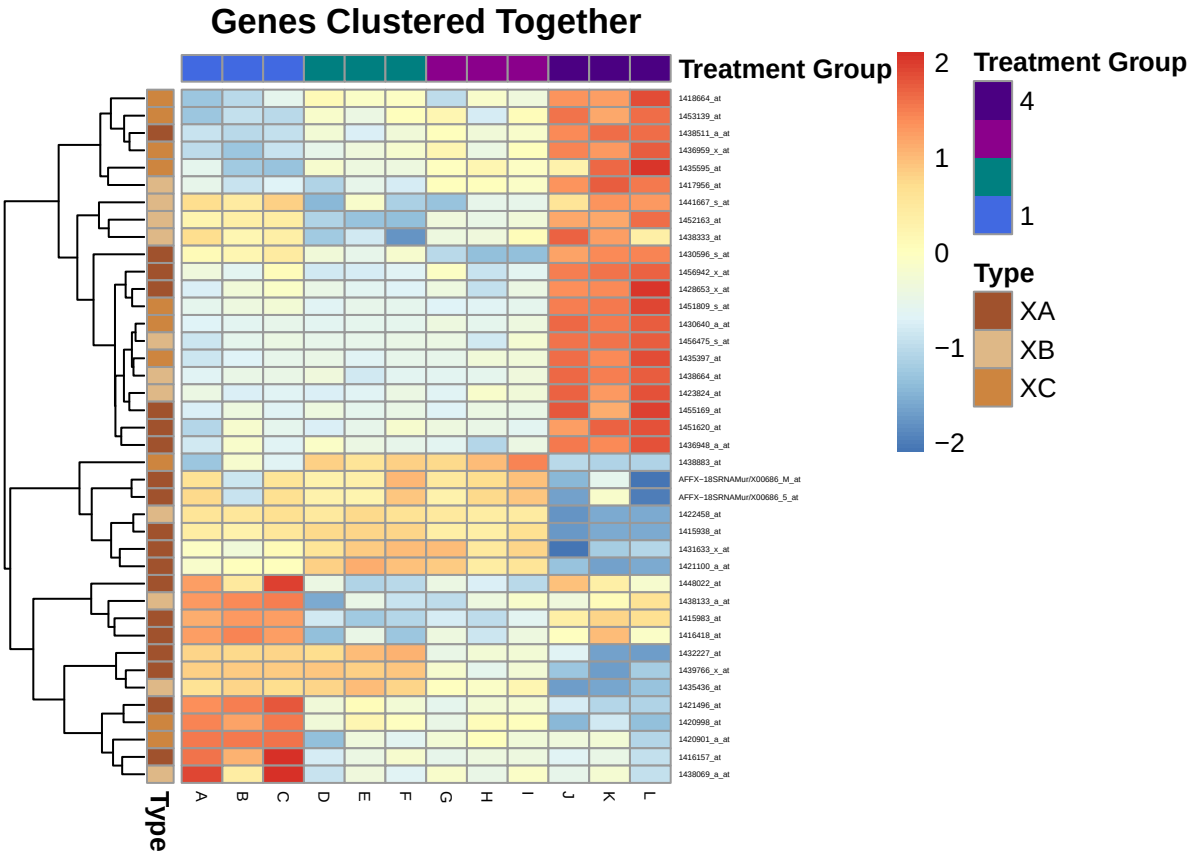
10. Using the annotations and table_bind data frame, a heatmap of expression analysis where both genes and samples are clustered was created.

```
#This chunk of code creates the heatmap where genes and samples are clustered together.
pheatmap(
  table_bind[1:12], scale='row',
  legend=TRUE, main = "Genes and Samples Clustered Together",
  annotation_row = rows_heatmap,
  annotation_col = cols_heatmap,
  annotation_colors = annotation_list,
  fontsize_row = 3,fontsize_col = 6)
```



10. Using the annotations and table_bind data frame, a heatmap of expression analysis where only genes are clustered was created.

```
#This chunk of code creates the heatmap where only genes are clustered together.
pheatmap(
  table_bind[1:12], scale='row',
  cluster_cols= FALSE, cluster_rows= TRUE,
  annotation_row = rows_heatmap,
  annotation_col = cols_heatmap,
  annotation_colors = annotation_list,
  legend=TRUE, main = "Genes Clustered Together",
  fontsize_row = 3,fontsize_col = 6)
```


Genes Clustered Together
Treatment Group
Type
A
B
C
D
E
F
G
H
I
J
K
L
2
1
0
−1
−2
Treatment Group
4
1
Type
XA
XB
XC